

ATHINAI ELEFThERIOS VENIZELOS, Greece

WMO: 167161

Lat: 37.936N

Lon: 23.944E

Elev: 308

StdP: 14.53

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	32.2	35.5	21.1	16.0	40.0	24.6	18.9	40.7	31.9	50.3	28.2	50.0	7.2	280	0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.1	95.0	68.6	91.8	68.0	89.9	67.4	72.0	86.3	70.7	85.5	69.7	85.0	13.5	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.7	103.0	78.6	66.0	97.1	77.9	64.4	91.6	77.3	35.9	86.3	34.9	85.3	34.0	84.5	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
27.2	24.5	22.1	DB	25.9	101.2	3.9	3.7	23.1	103.8	20.8	106.0	18.6	108.0	15.7	110.7
			WB	24.4	74.3	3.8	1.3	21.7	75.2	19.4	76.0	17.3	76.8	14.5	77.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	63.4	48.2	49.5	53.4	58.9	66.9	75.3	80.4	80.9	73.3	64.8	57.6	50.9	(6)
(7)		DBStd	12.54	5.99	6.11	4.90	4.74	4.76	5.19	3.54	3.41	4.51	4.84	5.58	5.89	(7)
(8)		HDD50	274	106	76	23	2	0	0	0	0	0	0	7	60	(8)
(9)		HDD65	2267	522	435	359	189	34	0	0	0	1	62	228	437	(9)
(10)		CDD50	5176	49	61	129	269	525	760	943	957	699	460	236	88	(10)
(11)		CDD65	1696	0	0	0	6	94	311	478	492	250	57	7	1	(11)
(12)		CDH74	18091	0	0	0	72	784	3326	5773	5869	2007	248	12	0	(12)
(13)		CDH80	7273	0	0	0	6	153	1301	2593	2605	593	21	1	0	(13)
(14)	Wind	WSAvg	8.8	8.5	9.0	8.8	8.2	7.7	8.5	10.5	11.2	9.0	8.6	7.8	8.0	(14)
(15)	Precipitation	PrecAvg	17.1	2.3	2.2	1.7	0.9	0.6	0.4	0.2	0.2	1.1	1.9	2.7	3.0	(15)
(16)		PrecMax	24.9	5.6	5.4	3.9	2.2	2.8	0.9	0.8	1.3	4.6	3.9	7.6	7.6	(16)
(17)		PrecMin	12.0	0.1	0.7	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.2	(17)
(18)		PrecStd	3.3	1.3	1.5	0.9	0.7	0.6	0.3	0.2	0.3	1.2	1.3	2.0	2.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.8	66.6	70.2	80.5	87.8	98.8	100.0	97.0	92.8	82.6	75.4	68.4	(19)
(20)		MCWB	55.1	55.6	56.2	58.7	64.2	68.2	68.9	68.8	67.7	65.4	63.1	59.6		(20)
(21)		2%	DB	62.6	64.4	68.0	75.5	84.1	92.8	94.9	94.7	88.1	79.1	71.9	66.2	(21)
(22)		MCWB	54.2	54.7	55.3	58.6	62.8	67.9	69.0	68.2	66.5	63.9	62.0	57.7		(22)
(23)		5%	DB	60.7	62.3	65.9	73.0	80.9	89.6	91.8	91.8	85.9	76.8	69.9	64.1	(23)
(24)		MCWB	53.3	53.9	54.4	57.4	61.6	66.9	68.0	68.2	65.7	63.3	61.4	56.3		(24)
(25)		10%	DB	58.8	59.4	63.0	69.8	78.6	87.4	89.8	89.9	83.8	73.8	67.7	61.0	(25)
(26)		MCWB	52.0	52.3	53.2	56.4	60.8	66.2	67.3	67.8	65.3	62.3	59.6	54.4		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.5	58.0	59.1	62.1	66.4	71.3	73.1	73.9	71.9	69.0	65.5	62.2	(27)
(28)		MCDB	62.9	65.0	67.0	73.9	80.9	88.8	87.5	87.9	83.2	77.2	73.1	67.8		(28)
(29)		2%	WB	55.5	56.1	57.0	60.2	64.8	69.9	71.6	72.5	70.1	67.2	63.6	59.0	(29)
(30)		MCDB	60.6	62.1	64.7	71.1	78.7	87.1	86.7	86.6	80.8	75.1	70.4	64.5		(30)
(31)		5%	WB	54.1	54.7	55.5	58.7	63.6	68.6	70.3	70.9	68.8	65.7	62.1	57.1	(31)
(32)		MCDB	59.6	60.4	62.9	69.5	76.8	85.3	86.2	85.6	79.7	73.7	68.4	62.8		(32)
(33)		10%	WB	52.2	53.0	54.3	57.3	62.4	67.5	69.4	69.8	67.5	64.0	60.4	55.1	(33)
(34)		MCDB	57.8	58.8	61.9	67.7	74.8	83.5	85.7	85.2	79.0	71.8	66.4	60.1		(34)
(35)	Mean Daily Temperature Range	MDBR	13.9	14.0	15.5	18.1	19.7	19.5	18.0	17.1	17.0	15.3	14.8	14.0		(35)
(36)		5% DB	MCDBR	16.3	17.5	19.7	24.0	24.1	22.7	21.9	20.8	21.8	19.4	16.6	16.2	(36)
(37)		MCWBR	11.0	11.1	11.2	12.0	9.9	8.4	8.4	8.4	9.5	9.7	10.1	10.8		(37)
(38)		5% WB	MCDBR	14.1	15.7	17.4	20.9	21.0	20.8	19.9	18.9	19.3	17.5	14.7	14.5	(38)
(39)		MCWBR	10.2	10.7	10.9	11.5	9.5	8.6	8.5	8.7	9.3	9.6	9.4	10.3		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.350	0.374	0.409	0.450	0.432	0.411	0.414	0.420	0.419	0.395	0.375	0.360		(40)
(41)		taud	2.431	2.347	2.247	2.138	2.254	2.349	2.347	2.323	2.328	2.370	2.391	2.405		(41)
(42)		Ebn at Noon	261	268	269	264	271	276	274	270	263	258	248	247		(42)
(43)		Edh at Noon	27	33	41	48	43	39	39	39	37	32	28	26		(43)
(44)	All-Sky Solar Radiation	RadAvg	628	875	1321	1741	2085	2372	2406	2145	1650	1122	746	546		(44)
(45)		RadStd	69	103	104	149	101	117	52	43	84	87	71	72		(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.63	N/A	N/A	N/A	N/A	N/A	-148	N/A	N/A	N/A

Nomenclature: See separate page